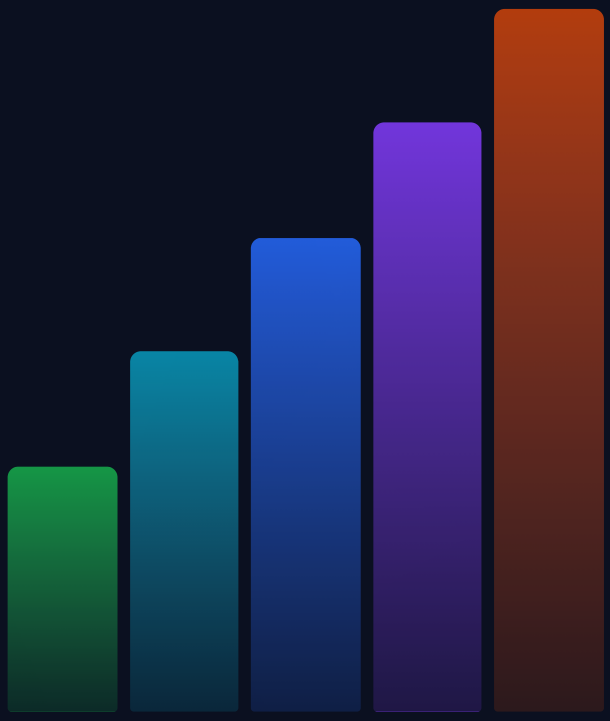
FIELD GUIDE · NOV 2026 → JUN 2028



Tier-3 to Top Offer

The placement field guide for a B.Tech CSE (Cybersecurity) student: DSA, Core CS, system design, security, AWS and AI, sorted into five difficulty levels, with a complete efficiency system (day modes, a 7-step problem routine, revision schedule, AI prompts, OA tactics), an off-campus hiring machine, and 16 real stories from students who did it from tier-3 colleges.

L0 · Foundation | L1 · Core | L2 · Interview-ready | L3 · Advanced | L4 · Elite

BUILT FROM
your Playbook, Roadmap and Cheat Sheet, plus public interview experiences, free courses and hiring pages

TAP ANY TOPIC
Underlined names open an in-depth page in the Placement Knowledge Base: concept, why it matters for you, code, mistakes, interview questions, practice

TRACK
Intensive: Striver core done by 30 Jan 2027 · AWS SAA by April · sprint 31 May–31 Jul · placements Aug–Dec 2027

START HERE

00 How this guide works

Every topic is a ladder. Find your rung, climb one at a time, and stop climbing where the target job stops asking.

L0
Foundation
You can't do it yet. Learn the syntax, the idea, the vocabulary. Goal: no blank page.

L1
Core
You can do it with notes. Standard problems, standard questions. Service-company OA level.

L2
Interview-ready
You can do it cold, timed, out loud. Product-company interview level (Amazon, Razorpay, Atlassian).

L3
Advanced
You can do the twist. Hards, contests, design trade-offs. Google, Microsoft, D. E. Shaw level.

L4
Elite
You do what few freshers do. CP rating 1900+, CVEs, open source, HFT puzzles. The outlier offers.

Which level each target needs

TARGET	DSA	CORE CS	DESIGN	SECURITY	CLOUD/AI
Mass recruiters (TCS Digital, Infosys SP, Accenture)	L1 · Core	L1 · Core	—	—	—
Product mid-tier (Razorpay, Zomato, Agoda, Myntra)	L2 · Interview-ready	L2 · Interview-ready	L1 · Core LLD	—	L1 · Core
Big Tech (Amazon, Microsoft, Google, Atlassian)	L3 · Advanced	L2 · Interview-ready	L2 · Interview-ready	—	L1 · Core
Security roles (AppSec, cloud security, SOC+)	L2 · Interview-ready	L2 · Interview-ready	L1 · Core	L3 · Advanced	L2 · Interview-ready AWS
HFT / quant dev (Tower, Graviton, Quadeye)	L4 · Elite	L3 · Advanced	L2 · Interview-ready	—	maths L3 · Advanced

EVERY NAME IS A DOOR
Topic names with a dotted underline link to their own page in the Placement Knowledge Base: the concept in depth, why it matters for your plan, key points, code, common mistakes, interview questions to tick off, practice problems and free resources. 97 topics.

THE ONE RULE
Depth on one ladder beats breadth on five. Your main ladder is DSA to L3 · Advanced. Core CS to L2 · Interview-ready. Security to L3 · Advanced because it's your degree and your edge. Everything else stops at L1 · Core – L2 · Interview-ready unless a specific job needs more.

01 THE ROUTE

Nov 2026 → Jun 2028 on one page

The intensive track you chose, with the semester and exams built in.

16 Nov – 13 Dec 2026 **Full force**
C++ in 2 days, then Striver sheet 5–6/day in order, LeetCode daily, 1 Core CS subject, cyber 1 h, AWS 30 min

14 Dec – 30 Jan 2027 **Sheet core done**
College 1–2 days/week. Finish Striver core (30 Jan). AI track starts. TenantWall build: CloudTrail, metric filter, alarm, SNS

Feb – 21 Mar 2027 **Hard pass 1 + paper**
Hard problems part 1, CP31 + Codeforces Div 2/3, OS + DBMS + CN revision, TenantWall paper by 21 Mar

22 Mar – 7 Apr 2027 **Semester exams**
Exams first. DSA drops to 1 revision problem/day. Nothing new.

8 Apr – 30 May 2027 **Hard pass 2 + AWS**
SAA-C03 exam week of 19 Apr. Hard part 2, LLD, 2 big projects, binary-exploitation taster

31 May – 31 Jul 2027 **Placement sprint**
2 mocks/week, company-tagged LeetCode, OA simulations, resume v5, 25 applications/week off-campus

Aug – Dec 2027 **Placements**
On-campus drives + off-campus machine running in parallel. Keep 3 problems/day.

Jan – Jun 2028 **Convert / upgrade**
Internship-to-PPO or keep interviewing for a better offer until graduation

Internship windows you can still hit

WINDOW	WHAT OPENS	ACTION NOW
Nov 2026 – Feb 2027	Off-campus summer 2027 intern roles still open at startups and mid-tier product cos; Unstop competitions (Flipkart GRiD, Myntra HackerRamp, etc. run yearly)	Apply weekly from mid-Dec once 150+ problems are done
Feb – Apr 2027	GSoC contributor applications (usually Mar–Apr); LFX Mentorship spring cohort	Pick 1 org in Jan, make 3 merged PRs before applying
Jun – Sep 2027	Big Tech new-grad + 6-month intern roles for the 2028 batch	Have resume, 2 projects, 400 problems ready by 31 May
Year-round	Bug bounty programmes, CTFs (KPMG, CyberSecurity events), open-source issues	One CTF a month from Feb

NOC NOTE
Your college gives no NOC by default, but you said a good company internship changes that. Have the offer letter first, then ask. Remote internships and 6-month Jan–Jun 2028 roles avoid most attendance clashes.

LEARN FASTER

02 The efficiency system

Twelve methods with evidence behind them. Use all of them every week; they cost minutes and save months.

01
Retrieval first

Close the video and solve or explain from memory before checking. Testing yourself beats rereading for long-term memory (Roediger & Karpicke, 2006).

02
Spaced revision 1-3-7-21

Re-solve each problem you got wrong after 1, 3, 7 and 21 days. Keep the list in a sheet; it's your most valuable file.

03
The 20-40 rule

Stuck 20 min on Easy/Medium (40 on Hard)? Read only the hint or the approach line, then code it yourself. Never copy code.

04
Pattern, not problem

After each solve, write one line: 'trigger → pattern'. E.g. 'sorted + pair sum → two pointers'. 300 problems become ~25 patterns.

05
Interleave

Mix topics in revision days (a graph, a DP, a string). Blocked practice feels faster; interleaved practice sticks better.

06
Speak while you code

Solve 1 problem a day out loud as if in an interview. It's a separate skill and the one tier-3 candidates practise least.

07
Error log

Every wrong submission: the bug in one line (off-by-one, overflow, missed empty case). Read the log before each contest.

08
Upsolve

After every contest, solve the next unsolved problem with the editorial. That one problem teaches more than the whole contest.

09
Feynman for Core CS

Explain deadlock, normalisation or TCP handshake on a blank page in 5 lines. Can't? That's your revision list.

10
50/10 deep blocks

50 min phone-in-another-room, 10 min break. 6 blocks/day is a full-force day. Track blocks, not hours.

11
Sleep 7+ h

Memory consolidates during sleep; all-nighters before contests cost more than they give.

12
Weekly review (Sun, 30 min)

What did I solve, what did I fail, what moves next week? Adjust the plan instead of abandoning it.

A full-force day (16 Nov → 13 Dec)

BLOCK	TIME	WORK
1–2	09:30–11:30	Striver sheet: 3 new problems in order (C++)
3	11:45–12:35	LeetCode: 1 daily + 1 revision from the 1-3-7-21 list
4	14:00–14:50	Core CS: one subtopic, Feynman page, 10 MCQs
5	15:00–15:50	Striver: 2–3 more problems
6	17:00–18:00	Cyber: one PortSwigger / TryHackMe lab, timed
7	18:15–18:45	AWS 30 min (this week's module)
8	21:00–21:30	Speak-aloud problem + error log + tomorrow's 3 problems picked

02·A EFFICIENT PROCESS
Run your prep like a system

Decide once, not every morning. Pick today's mode, follow its recipe, and let the loops catch what slips.

Five day modes: pick one each morning in 10 seconds

MODE	WHEN	MINIMUM THAT COUNTS AS A WIN	DROP
Full force	Home, no college	6–8 deep blocks: 5–6 Striver, 1 LC, Core CS, cyber, AWS	nothing
College day	Attending (1–2/week)	2 problems on the commute/evening + AWS 30 min + 10 Core CS flashcards	cyber, new topics
Exam mode	22 Mar – 7 Apr	1 revision problem from the 1-3-7-21 list	everything else
Low energy	Tired, ill, bad day	1 Easy re-solve + read your error log (20 min)	the guilt
Sprint	31 May – 31 Jul	Mock or OA sim + company-tagged set + 5 applications	new learning

> **NEVER-ZERO RULE**
> The streak is "at least one problem touched", not "a perfect day". A low-energy day that keeps the streak is worth more than a perfect day followed by four days off.

Four loops that keep the plan honest

Daily · 5 min at night
Pick tomorrow's 3 problems and 1 Core CS topic. Write them down. Mornings start with no decisions.

Weekly · Sun 30 min
Count problems, contests, blocks. Re-solve the week's failed problems. Move unfinished items, never stack them.

Monthly · last Sun 1 h
Check the milestone box (p.19). One timed mock on old topics. Cut whatever isn't moving a number.

Phase · each milestone
Rewrite the resume, update LinkedIn/GitHub, adjust the next phase's daily quota up or down by one problem.

Decision rules: so you never waste an hour deciding

- **Stuck?** 20 min (Easy/Med) or 40 min (Hard) → read only the approach → code it yourself → add to the 1-3-7-21 list.
- **Video or article?** Article first (faster, searchable). Video only if the article didn't click, at 1.5×.
- **New resource tempting you?** Only if the current one has no material on that topic. Finish the sheet before switching.
- **Behind schedule?** Cut the newest track (AI, then HFT), never DSA or revision. Never double tomorrow's quota.
- **Two good options?** Pick the one that produces a proof (solved problem, lab, PR, resume line).
- **Feeling ready to apply?** You never will. Apply when the checklist says so (≈150 problems + 1 project).

EFFICIENT PROCESS

02·B One problem, done the efficient way

The same 7 steps for every problem. This turns 550 problems into skill instead of a number.

1
Read + restate (2 min)
Rewrite the task in one line. Note n's limit → target complexity (p.10 table). List 3 edge cases: empty, one element, max/duplicates/negatives.

2
Brute force first (3 min)
Say or write the simplest correct idea and its complexity. Interviewers reward this; it also gives you a checker.

3
Name the pattern (2 min)
Use the trigger table. If none fits, try: sort it? hash it? binary search the answer? think in states (DP)?

4
Code it (≤15 min)
Clean names, no premature tricks. C++ template below. Stop at the timebox and take the hint.

5
Dry run (3 min)
Trace one normal + one edge case by hand before submitting. Catches most WA.

6
Log it (1 min)
Sheet row: link · pattern · trigger line · mistake · next revision date. This is the most valuable minute.

7
Revise on schedule
Day 1, 3, 7, 21: re-solve from a blank file in ≤10 min. Pass twice in a row → retire it.

Tracker sheet columns (Google Sheets, free)

`Date · Problem · Link · Level · Pattern · Trigger · Time · Hint? · Mistake · R1 · R3 · R7 · R21`

Formula for next revision: `=A2+1`, `=A2+3`... then filter "due ≤ today" each morning. 10 minutes of setup saves hundreds of re-solves.

Reading an editorial efficiently

1. Read only until you see the key idea, then close it.
2. Code it yourself without looking.
3. Only then read their code and compare.
4. Write the one line you missed.

Contest C++ template (compile-checked)

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;
#define all(x) (x).begin(), (x).end()
const ll INF = 4e18; const int MOD = 1e9 + 7;
void solve() {
    int n; cin >> n;
    vector<ll> a(n);
    for (auto &x : a) cin >> x;
    // brute force first, then optimise
}
int main() {
    ios::sync_with_stdio(false); cin.tie(nullptr);
    int t = 1; cin >> t;
    while (t--) solve();
}
```

Online assessment and contest tactics

Read all problems first
2 minutes. Solve in order of confidence, not order given.

Partial scores count
HackerRank-style OAs score test cases. A correct brute force on half the tests beats a perfect idea not submitted.

Watch overflow
Sums/products of ints → `long long`. Most silent WA in OAs.

Custom tests
Run the edge cases you listed before submitting. Empty, 1, max n, all equal.

Last 10 minutes
Stop optimising. Submit what passes, add guards for edge cases.

Same environment
Practise on HackerRank/HackerEarth too; their I/O and editors differ from LeetCode.

02·C EFFICIENT PROCESS
Use AI, tools and the 80/20 rule

AI is a tutor and interviewer, never your hands. Tools remove friction. The 80/20 list tells you what to learn first.

Six prompts for Claude or Gemini

HINT-ONLY TUTOR

I'm solving [problem link/text]. Don't give the solution or code. Give me one hint at a time, the smallest nudge first. Wait for me to say 'next' before the next hint.

MOCK INTERVIEWER

Act as an Amazon SDE interviewer. Give me one Medium DSA problem. Stay silent while I think aloud, ask about edge cases and complexity, then score me 1–5 on problem solving, code and communication with one fix each.

CODE REVIEWER

Here is my accepted C++ solution. Don't rewrite it. Point out: bugs on edge cases, complexity, unclear names, and one cleaner STL idiom I could have used.

FEYNMAN CHECKER

I'll explain [deadlock / B+ tree / TCP handshake] in 5 lines. Find what's wrong or missing, then ask me 3 interview follow-up questions on it.

FLASHCARD MAKER

Turn these notes into 10 question–answer flashcards, one fact each, in 'Q; A' format for Anki import.

RESUME REVIEWER

Review this resume for an SDE intern role at a product company. For each bullet: is there a number? an action verb? a technology? Rewrite the weakest 3.

THE AI RULE

Never paste a problem and ask for the solution before your 20/40-minute timebox is up. Copying AI code feels like progress and gives zero interview skill — interviewers can tell in 2 minutes.

The 20% that gets 80% of the questions

AREA	LEARN THESE FIRST
DSA	Arrays + hashing, two pointers, sliding window, binary search, trees (traversals, BST, LCA), graphs (BFS/DFS, topo, Dijkstra), DP (1D, 2D grid, knapsack, LIS, LCS), heaps, stacks
OS	Process/thread, scheduling, deadlock, paging + virtual memory, synchronisation (mutex, semaphore)
DBMS + SQL	Normalisation, ACID + isolation, indexing, joins, GROUP BY + HAVING, window functions, Nth highest
CN	OSI/TCP-IP, TCP vs UDP, handshake, HTTP/HTTPS/TLS, DNS, 'type a URL' walkthrough
OOP	4 pillars, virtual/vtable, overloading vs overriding, SOLID, 3 patterns
Security	OWASP Top 10, auth vs authz, hashing vs encryption, TLS, IAM least privilege

Free tool stack

NEED	TOOL	WHY IT SAVES TIME
Tracker + revision dates	Google Sheets	Filter 'due today' in one click
Core CS memory	Anki	Spaced repetition done for you, 10 min/day
Focus blocks	Pomofocus + phone in another room	Counts blocks, not hours
Diagrams (LLD/HLD, TenantWall)	Excalidraw, diagrams.net (AWS icons)	Fast, exportable, looks pro
Contest calendar	clist.by	All contests in one place
Mock interviews	Pramp, peers, AI mock prompt	Free, real-time speaking practice
Job alerts	LinkedIn + Naukri + company career pages saved searches	Jobs come to you; apply within 48 h

02·D

EFFICIENT PROCESS

Apply, interview and last the distance

Batch the boring parts, reuse everything, and protect the energy that does the real work.

Application pipeline in 90 min/week

1. **Master resume** with every bullet you have (2 pages, private).
2. **3 role versions** cut from it: SDE, Security/AppSec, Cloud. One page each.
3. **Tailor in 5 min**: swap 2 bullets and the skills order to match the job's top 5 keywords. Nothing more.
4. **Batch**: Mon + Thu evening, 45 min each. Alerts collected, applications sent, referral messages sent, tracker updated.
5. **Follow up** once after 5–7 days, from the tracker's "next action" column.
6. **Reuse answers**: keep a doc of "Why this company / Tell me about yourself / Strengths" answers, edit per company.

Company-specific prep in 1 week

1. Read 5–10 recent interview experiences for that company and role (LeetCode Discuss, GfG).
2. Solve its last-6-months LeetCode tag list, top 30 by frequency.
3. Map its values (e.g. Amazon Leadership Principles) to your 6 STAR stories.
4. Know its product: use it, find one thing you'd improve, one security thought.

Interview day: efficient habits

- Night before: no new topics. Re-read your error log, pattern sheet and STAR stories.
- Have a 60-second "tell me about yourself": who you are, 2 proofs (rating, TenantWall), why this role.
- Think aloud from second one; silence is scored as stuck.
- State brute force, then improve. Ask before coding: "Shall I code this approach?"
- End every round with one smart question about the team's work.
- After: write down every question within 1 hour. That list is your next week's revision.

Energy: the hidden multiplier

- **Sleep 7–8 h** at a fixed time; it's when memory consolidates.
- **Hardest work first** (new DSA) in your sharpest hours; revision and AWS when tired.
- **Move 20–30 min** daily (walk, cricket, gym). Improves focus more than coffee.
- **Phone out of the room** during blocks; social apps only after block 4.
- **One rest half-day a week.** Burnout costs weeks; rest costs hours.
- **Compare with your last month**, not with LinkedIn posts.

> **EFFICIENCY IN ONE LINE**
> Fixed modes, a 7-step problem routine, a revision sheet that tells you what to redo, AI as a tutor not a solver, batched applications, and sleep. Everything else in this guide plugs into those six.

MAIN LADDER

03 DSA from L0 to L4

The ladder that decides most offers. Climb it in order; don't skip L2 to chase L4.

L0 FOUNDATION

C++ and STL in 2 days

vector, string, pair, map, unordered_map, set, priority_queue, stack, queue, deque; sort with lambdas; iterators; `auto`; fast I/O. Big-O of every operation.

Use: takeuforward STL, your Cheat Sheet §3, learncpp.com (chapters you need)

Done when: you can write any STL container op without searching

L1 CORE

Striver A2Z: steps 1–9

Basics, sorting, arrays, binary search, strings, linked list, recursion, bit manipulation, stack/queue. Easy + Medium in sheet order, 5–6 a day.

Use: Striver A2Z sheet, LeetCode daily

Done when: you solve 80% of step problems within 25 min without hints

L2 INTERVIEW-READY

Striver steps 10–18 + patterns

Sliding window, two pointers, heaps, greedy, binary trees, BST, graphs (BFS/DFS, Dijkstra, topo sort, DSU), DP 1D/2D/subsequence/LIS/strings, tries.

Use: Striver sheet, NeetCode 150 as a cross-check, LeetCode company tags

Done when: LC Medium in ≤25 min, out loud, with complexity — 70% of the time

L3 ADVANCED

Hard problems + contests

Hards from the sheet in 2 passes; monotonic stack, interval DP, bitmask DP, DP on trees, segment tree / BIT, string hashing, KMP/Z. Weekly LC contest + Codeforces Div 2.

Use: CSES, CP31 sheet, Codeforces, cp-algorithms

Done when: LC contest rating 1800+ or CF 1400+; LC Hard in 45 min half the time

L4 ELITE

Competitive edge

CF Div 1/2 D–E, advanced graphs (flows, SCC, bridges), heavy-light, centroid, matrix exponentiation, probability DP, game theory. ICPC-style team practice.

Use: CSES advanced sections, USACO Guide gold/platinum, CF gym, AtCoder ABC/ARC

Done when: CF 1900+ / LC Knight-Guardian; needed only for HFT/quant-dev or top-tier Google

Problem counts that are enough

BY	TOTAL SOLVED	MIX	SIGNAL
30 Jan 2027	≈ 300	Striver core E/M	L1 · Core → L2 · Interview-ready
30 May 2027	≈ 450	+ 120 Hards/tagged	L2 · Interview-ready → L3 · Advanced
31 Jul 2027	≈ 550	+ company-tagged, mocks	L3 · Advanced

04 PATTERN TRIGGERS

Read the problem, name the pattern

20 triggers that cover most interview problems. Cover the right column and quiz yourself.

IF THE PROBLEM SAYS…	REACH FOR	LEVEL
Sorted array, find pair/triple	**Two pointers**	L1 · Core
Longest/shortest subarray with condition	**Sliding window**	L1 · Core
Min/max answer that is monotone	**Binary search on answer**	L2 · Interview-ready
Next greater / smaller	**Monotonic stack**	L2 · Interview-ready
k largest / streaming median	**Heap(s)**	L2 · Interview-ready
Subarray sum = k	**Prefix sum + hashmap**	L1 · Core
All subsets / permutations	**Backtracking**	L1 · Core
Shortest path, unweighted	**BFS**	L1 · Core
Shortest path, weighted ≥0	**Dijkstra**	L2 · Interview-ready
Dependencies / order	**Topological sort**	L2 · Interview-ready
Connected groups, merging	**DSU**	L2 · Interview-ready
Count ways / min cost over choices	**DP**	L2 · Interview-ready
Two strings compare	**2D DP (LCS/edit)**	L2 · Interview-ready
Prefix matching many words	**Trie**	L2 · Interview-ready
Range query + update	**Segment tree / BIT**	L3 · Advanced
Intervals overlap	**Sort + sweep**	L2 · Interview-ready
n ≤ 20, choose subsets	**Bitmask DP**	L3 · Advanced
Tree answer per node	**DFS + DP on trees**	L3 · Advanced
Max XOR pair	**Bit trie**	L3 · Advanced
Pattern in text, O(n)	**KMP / Z / hashing**	L3 · Advanced

Constraint → complexity

N UP TO	10	20	500	5 000	10^6	10^9+
Aim for	O(n!)	$O(2^n)$	$O(n^3)$	$O(n^2)$	O(n log n)	O(log n) / O(1)

THE INTERVIEW LOOP (UMPIRE)

Understand (repeat it, ask about constraints and edge cases) · **M**atch a pattern · **P**lan in words, state complexity · **I**mplement cleanly · **R**eview with a dry run · **E**valuate trade-offs and a better approach. Talk the whole time.

05 CORE CS
OS, DBMS, CN, OOP, SQL by level

Interviewers ask the same 40 questions. Know them at L2 and you've covered most rounds.

SUBJECT	L1 · Core KNOW	L2 · Interview-ready EXPLAIN + EXAMPLE	L3 · Advanced DEPTH
OS	process vs thread, scheduling algos, deadlock conditions	paging, virtual memory, page faults, mutex vs semaphore, producer-consumer, thrashing	TLB, copy-on-write, context-switch cost, Linux scheduler (CFS), memory-mapped files
DBMS	keys, ER, normal forms to 3NF, ACID	BCNF, indexing (B+ tree), transactions and isolation levels, deadlock, joins	MVCC, write-ahead logging, query plans, sharding vs replication, CAP
CN	OSI vs TCP/IP, TCP vs UDP, IP classes, DNS	3-way handshake, flow vs congestion control, HTTP/HTTPS + TLS, subnetting, what happens when you type a URL	TCP congestion algorithms, QUIC/HTTP3, BGP basics, NAT traversal, CDN
OOP	4 pillars with C++ examples	virtual functions + vtable, overloading vs overriding, abstract vs interface, SOLID	design patterns (singleton, factory, observer, strategy), RAII, rule of 5
SQL	SELECT, WHERE, GROUP BY, HAVING, JOIN	subqueries, window functions (RANK, ROW_NUMBER, LAG), Nth highest salary	query optimisation, indexes in practice, CTE recursion

Use: GfG OS · Striver/takeuforward Core CS notes · Gate Smashers for weak topics · LeetCode SQL 50 · SQLBolt

The 12 most-asked questions (answer each in 60 seconds)

1. Process vs thread — and why threads are cheaper
2. Four conditions of deadlock and how to prevent one
3. What is virtual memory and why do we need it?
4. Mutex vs semaphore with an example
5. Explain normalisation up to 3NF using one table
6. What are ACID properties? Which isolation level prevents dirty reads?
7. How does a B+ tree index speed up a query?
8. What happens when you type google.com and press Enter?
9. TCP vs UDP — when would you choose UDP?
10. Explain the TCP 3-way handshake and why not 2-way
11. Runtime polymorphism in C++: how does the vtable work?
12. Write SQL for the second-highest salary per department

06 BUILD AND DESIGN
System design and projects

Freshers get LLD more than HLD. Projects get you shortlisted; design talk gets you hired.

L1 CORE

LLD basics

Classes, relationships, SOLID, 4 patterns. Design a parking lot, library system, Splitwise, elevator in C++.

Use: awesome-low-level-design (free)

Done when: you can design a parking lot with classes in 30 min

L2 INTERVIEW-READY

HLD fundamentals

Load balancer, caching, CDN, DB choice (SQL vs NoSQL), replication, sharding, message queues, rate limiting. Design URL shortener, chat app, news feed.

Use: system-design-primer, roadmap.sh, your AWS SAA notes

Done when: you can draw and defend a URL shortener with numbers

L3 ADVANCED

Trade-off depth

Consistency models, CAP/PACELC, idempotency, distributed IDs, back-pressure, observability.

Use: Designing Data-Intensive Applications (Kleppmann) chapters 1–9; engineering blogs (Razorpay, Zomato, Uber)

Done when: you can argue two designs and pick one with reasons

Projects that work for a cyber + SDE profile

PROJECT	SHOWS	LEVEL
TenantWall: ABAC multi-tenant isolation on AWS with CloudTrail, metric-filter alarm, SNS, Access Analyzer + paper	cloud security, IAM depth, research	L3 · Advanced
Rate-limited URL shortener (C++/Go or Node) with Redis cache, load test numbers, deployed on AWS	backend + system design	L2 · Interview-ready
Log-based intrusion detector: parse auth/nginx logs, flag brute force and scans, dashboard	security + data handling	L2 · Interview-ready
Secure AI assistant: RAG over your notes with prompt-injection tests and guardrails	AI + AppSec (rare combo)	L3 · Advanced
Open-source PRs (3+ merged) in one org	real code review, collaboration	L3 · Advanced

PROJECT RULE

Every project needs: a live link or demo video, a README with an architecture diagram, one number (latency, users, requests/s, bugs found), and a line on what you'd do next. Two great projects beat six tutorials.

07 YOUR EDGE
Security ladder

Your degree is cybersecurity. Most SDE candidates can't talk about it — use that in every interview.

L0 FOUNDATION

Foundations

Linux command line, networking basics, HTTP, how the web works.

Use: OverTheWire Bandit, TryHackMe Pre-Security (free rooms)

Done when: Bandit level 20+

L1 CORE

Web security

OWASP Top 10: SQLi, XSS, CSRF, SSRF, IDOR, auth flaws, access control.

Use: PortSwigger Web Security Academy (free)

Done when: all Apprentice labs + 30 Practitioner labs

L2 INTERVIEW-READY

Practical AppSec + cloud

Burp Suite workflow, code review for vulns, secure coding in C++, AWS IAM/S3 misconfigurations, threat modelling (STRIDE).

Use: PortSwigger Practitioner, flaws.cloud, OWASP

Done when: you can explain and fix 5 vulns in code

L3 ADVANCED

Binary + CTFs

C memory, stack layout, buffer overflow, format strings, ROP basics, GDB/pwndbg, Ghidra reversing. Jeopardy CTFs monthly.

Use: pwn.college, picoCTF, CTFtime

Done when: top-25% finish in a mid CTF; 3 write-ups on a blog

L4 ELITE

Proof in the wild

Valid bug bounty reports, a CVE, a security tool on GitHub with users, conference/college talk.

Use: HackerOne, Bugcrowd, vendor disclosure programmes

Done when: one accepted report or CVE

USE IT IN SDE INTERVIEWS

When asked about any project: "I also threat-modelled it — here's the IDOR I found and fixed." When asked about databases: mention SQL injection and parameterised queries. One security sentence per answer makes you memorable.

MULTIPLIERS

08 AWS, AI and the HFT stretch

Each one adds a line to your resume that most tier-3 candidates don't have.

TRACK	L1 · Core	L2 · Interview-ready	L3 · Advanced
AWS (30 min daily)	IAM, EC2, S3, VPC basics; billing alarm	AWS Academy Cloud Architecting done; SAA-C03 passed (Apr 2027)	TenantWall in production-style; security specialty later
AI (≤1 h from 14 Dec)	Python, NumPy, how LLMs work, prompting	Build RAG + tool-calling app; evals	Prompt-injection testing, AI security (OWASP LLM Top 10)
HFT maths (Apr–Jul 2027)	Counting, Bayes, expectation	Green-book probability, Brainstellar puzzles, Zetamac 40+	Market-making games, CF 1600+, low-latency C++

Use: AWS Academy (via your faculty) · freeCodeCamp SAA course · DeepLearning.AI short courses · OWASP LLM Top 10 · Stat 110 · Brainstellar · Zetamac

Resume: one page, ATS-safe

Order

1. Name, phone, email, LinkedIn, GitHub, LeetCode/CF links
2. Education (CGPA if ≥ 7.5)
3. Experience / internships
4. Projects (2–3, each 2–3 bullets)
5. Achievements (ratings, CTF ranks, SAA cert, paper)
6. Skills (languages, tools, cloud)

Bullet formula

`Did X using Y, which resulted in Z (number)`

Weak: Made a URL shortener using Node.

Strong: Built a URL shortener serving 1.2k req/s on one t3.micro by adding Redis caching and a token-bucket rate limiter; p95 latency 18 ms.

Single column, no photos, no tables, PDF, file name `Firstname_Lastname_SDE.pdf`.

09 OFF-CAMPUS

The hiring machine

On-campus is a lottery you can't control. Off-campus is a numbers game you can.

CHANNEL	HOW	WEEKLY QUOTA
Referrals	Find alumni/employees on LinkedIn at target companies; send the referral note below with job ID + resume	10 messages
Careers pages	Bookmark 40 companies' careers pages; check Mon + Thu. Apply within 48 h of posting	10 applications
Hiring contests	Unstop, HackerEarth, Naukri Campus, company hackathons (Flipkart GRiD, Myntra HackerRamp, Amazon ML Summer School, JPMC Code for Good)	every one that opens
Cold email	Startup founders/CTOs (YC India, funded on Inc42/Entrackr) with a 5-line email + one relevant project	5 emails
Open source	GSoC, LFX Mentorship, MLH Fellowship; maintainers refer contributors	2 PRs
Contest ratings	Recruiters filter on LC/CF rating; some firms invite top contest rankers	1–2 contests

Referral message (LinkedIn, 300 chars)

Hi [Name], I'm a 3rd-year CSE (Cybersecurity) student who has solved 450+ DSA problems and built [project — one line]. I'm applying for [Role, Job ID] at [Company]. Would you be open to referring me? Resume: [link]. Happy to share anything else. Thank you!

Cold email to a startup

Subject: SDE intern — built [X] relevant to [their product]

Hi [Name], I use/studied [product]. I'm a CSE (Cyber) student; I built [project + number], and I noticed [specific thing — a bug, a security header missing, a feature idea]. I'd love to intern with your team for [months]. Resume + GitHub: [links]. Could we talk for 15 minutes this week?

TRACK IT

One sheet: company, role, date, channel, contact, status, next follow-up. Follow up once after 5–7 days. Expect ~2–5% replies; 50 good applications a month is what produces interviews.

10 THE ROOM

Interviews, rounds and offers

What each round checks, and how to prepare for it.

ROUND	WHAT IT CHECKS	PREPARE BY
Online assessment (60–90 min)	2–3 DSA problems, sometimes MCQs on CS/aptitude	Timed LC contests, HackerRank practice, read every line of constraints
DSA interview (45–60 min)	1–2 problems, communication, edge cases, complexity	Speak-aloud practice, 2 mocks/week from June (Pramp, peers)
Core CS / technical	OS, DBMS, CN, OOP, SQL, your projects in depth	The 12 questions, Feynman pages, project deep-dive script
LLD / HLD (product cos)	Structuring classes, trade-offs	6 LLD designs written in C++, 4 HLD diagrams
Behavioural / bar raiser	Ownership, conflict, failure, learning	6 STAR stories that each fit several questions
HR	Fit, relocation, notice, compensation	Research the company; know your number

Six STAR stories to prepare

1. A hard technical problem you solved (TenantWall IAM)
2. A time you failed and what you changed
3. A conflict in a team
4. Learning something fast (C++/AWS in weeks)
5. Going beyond the task (security finding in a project)
6. Leading or helping someone

Offer basics

- CTC ≠ in-hand. Ask for the split: fixed, variable, joining bonus, stocks (vesting).
- Never say a number first if you can avoid it; if pushed, give a range from levels.fyi / Glassdoor.
- Check the bond/service agreement clause before signing.
- A written offer + another interview in progress is your leverage — be polite and honest.

Mock schedule (from 31 May)

MON	TUE	WED	THU	FRI	SAT	SUN
DSA mock (peer)	Company-tagged set	Core CS mock	LLD design	OA simulation 90 min	LC + CF contest	Review + STAR practice

11

THEY DID IT FROM TIER-3

Stories

SDE offers and internships, from public interview experiences.

01 FULL-TIME · 2024 · SELF-REPORTED

Tier-3 → Microsoft, Google and Meta

Rejected by Google 3 times and Meta twice before getting selected at both.

Copy this: Treat rejection as data: log which round failed and fix that round only. Persistence over several cycles is normal, not a sign you're not good enough.

LeetCode Discuss

02 INTERNSHIP OFFER · AUG 2025 · SELF-REPORTED

Tier-3 → Razorpay SDE intern

Came through the college T&P cell; the drive had an 8+ CGPA cut-off.

Copy this: CGPA still opens doors at fintechs. Keep yours above 8 if you can — it's cheaper than any course.

LeetCode Discuss

03 OFF-CAMPUS · DEC 2025 · SELF-REPORTED

Tier-3 → Eternal (Zomato) SDE intern + FTE

Got in through a referral and cleared the intern + full-time process.

Copy this: Referrals are the strongest off-campus channel. Ten referral messages a week is your quota.

LeetCode Discuss

04 OFFER · MAY 2026 · SELF-REPORTED

Agoda SDE intern, 6 months, Gurugram

A six-month SDE internship offer, documented round by round.

Copy this: Six-month internships (Jan–Jun of final year) are a real route to a PPO and avoid the summer NOC problem.

LeetCode Discuss

05 SELECTED · SELF-REPORTED

Amazon SDE intern, off-campus

An off-campus Amazon internship: online assessment, then DSA interviews.

Copy this: Amazon tests DSA + Leadership Principles. Pair every DSA mock with one STAR story.

GeeksforGeeks

06 INTERNSHIP · SELECTED · SELF-REPORTED

Myntra RampUp, via Unstop

Entered through an Unstop hiring programme rather than a referral.

Copy this: Hiring contests on Unstop skip the resume filter. Enter every relevant one.

Medium

07 SELECTED · SELF-REPORTED

INDmoney SDE intern, off-campus

A fintech startup internship with DSA and project rounds.

Copy this: Startups ask about your projects in depth — have numbers ready.

GeeksforGeeks

08 SELECTED · SELF-REPORTED

Scaler SDE intern, off-campus

Ed-tech internship through an off-campus process.

Copy this: Mid-size companies are reachable earlier — use them as interview practice that might also be an offer.

GeeksforGeeks

WHAT EVERY STORY HAS IN COMMON

(1) Hundreds of problems solved before the first offer. (2) One channel that beat the college filter: a referral, a contest, a CTF, open source. (3) At least one rejection first. The college name decided none of them.

12 THEY DID IT FROM TIER-3

Stories (part 2)

More routes — including security, open source and the honest failures.

09 FTE + INTERN · OFF-CAMPUS · JAN 2022 · SELF-REPORTED

Accolite Digital, Mumbai

An off-campus offer from a Mumbai engineering services company.

Copy this: Mumbai-based companies are worth a separate list — no relocation question, and local alumni refer more readily.

LeetCode Discuss

10 FULL-TIME · SELF-REPORTED

Tier-3 → tier-1 software company

A personal account of moving from a tier-3 college to a top product company.

Copy this: Read the whole story for the timeline: the gap between 'started' and 'offer' is usually 12–24 months. You're on that clock now.

Medium

11 OPEN SOURCE · SELF-REPORTED

Tier-3 MCA → Google Summer of Code 2026

Got into GSoC 2026 as a tier-3 MCA student.

Copy this: GSoC is a brand that removes the college question. Start contributing in Jan; applications are usually Mar–Apr.

DEV

12 SECURITY INTERNSHIP · COMPANY PAGE

B.Tech → KPMG India cyber (CDIR) intern via CTF

A KPMG campus CTF was central to moving from final-year B.Tech student to intern on KPMG India's cyber incident response team.

Copy this: CTF results are a hiring channel for security roles. Play one a month from Feb and write up what you solve.

KPMG India

13 SECURITY CAREER · 2015 · INTERVIEW (OLDER)

Bug bounties → security engineer

Anand Prakash started reporting bugs to Facebook and others, which led to a security engineering career.

Copy this: Accepted bug reports are proof no college name can match. One valid report is worth a line at the top of your resume.

YourStory

14 CONTRACT ROLE · NEWS REPORT

Reported CBSE portal flaws → IIT Kanpur cyber team

A teenager who reported security flaws in CBSE's marks portal was taken on as an engineer in IIT Kanpur's cybersecurity team (contractual).

Copy this: Responsible disclosure, written up publicly, can be a résumé on its own.

Siasat

15 SELF-TAUGHT CAREER · COMPANY BLOG

Biology graduate → cybersecurity

Anugrah S.R. moved from biology into cybersecurity through self-learning.

Copy this: If a non-CS graduate can do it, a CSE (Cyber) student with a plan is already ahead.

CloudSEK

16 NOT SELECTED · SELF-REPORTED

Two honest rejections: Microsoft and Cisco

Two tier-3 candidates wrote up interviews they failed.

Copy this: Read failures as carefully as wins: most write-ups point to communication, edge cases or weak Core CS — all fixable before Aug 2027.

Medium + LeetCode

HOW TO READ THESE

Most are self-reported posts; outcomes aren't independently checked. The Cisco rejection is at LeetCode Discuss. Aggregator pages that list tier-3 success stories without sources (for example jobrise.io) are left out of the cards. I found no published tier-3 stories for security-engineer full-time offers specifically — the security routes above are CTFs, bug bounties and disclosure.

13 CHECK YOURSELF

Milestone checklists

Tick these on paper. If a box is empty at its date, the next month fixes that box first.

13 Dec 2026

- [ ] C++ STL fluent
- [ ] Striver steps 1–6 done (~150 problems)
- [ ] OS notes complete
- [ ] PortSwigger Apprentice labs done
- [ ] AWS weeks 1–4 + billing alarm
- [ ] Error log + 1-3-7-21 sheet in use

30 Jan 2027

- [ ] Striver core done (~300)
- [ ] LC contest every week
- [ ] DBMS + SQL 50 done
- [ ] TenantWall: trail, metric filter, alarm, SNS live
- [ ] AI: first RAG demo
- [ ] Resume v1

21 Mar 2027

- [ ] Hard pass 1 done
- [ ] CF Div 2: solve A–C regularly
- [ ] CN + OOP notes complete
- [ ] TenantWall paper submitted
- [ ] 1 CTF write-up
- [ ] 20 off-campus applications sent

30 May 2027

- [ ] ~450 problems, 120 Hards
- [ ] AWS SAA-C03 passed
- [ ] 4 LLD + 2 HLD designs
- [ ] 2 deployed projects with numbers
- [ ] Binary taster: 5 pwn.college levels
- [ ] Resume v4 reviewed by 2 seniors

31 Jul 2027

- [ ] ~550 problems, LC 1800+
- [ ] 16 mocks done
- [ ] 6 STAR stories rehearsed
- [ ] Company-tagged lists for top 15 targets
- [ ] 100 applications + 40 referral asks
- [ ] LinkedIn + GitHub cleaned up

Any week

- [ ] ≥ 30 problems solved
- [ ] ≥ 1 contest
- [ ] ≥ 3 h Core CS
- [ ] ≥ 5 h cyber
- [ ] ≥ 3.5 h AWS
- [ ] Sunday review done

MISTAKES THAT COST TIER-3 STUDENTS OFFERS

Watching solutions instead of solving · 1000 problems with no revision · no speaking practice · projects copied from tutorials · applying only on campus · waiting to "be ready" before applying · ignoring CGPA · quitting after the first rejection.

14 SOURCES

Everything linked in this guide

All free unless noted. Story links are in sections 11–12.

Courses and practice

Striver A2Z DSA sheet
https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2/

NeetCode practice
https://neetcode.io/practice

CSES problem set
https://cses.fi/problemset/

cp-algorithms
https://cp-algorithms.com

USACO Guide
https://usaco.guide

roadmap.sh
https://roadmap.sh

LeetCode SQL 50
https://leetcode.com/studyplan/top-sql-50/

System Design Primer
https://github.com/donnemartin/system-design-primer

Awesome LLD
https://github.com/ashishps1/awesome-low-level-design

PortSwigger Academy
https://portswigger.net/web-security

pwn.college
https://pwn.college

OverTheWire
https://overthewire.org/wargames/bandit/

picoCTF
https://picoctf.org

CTFtime
https://ctftime.org

flaws.cloud
https://flaws.cloud

OWASP LLM Top 10
https://genai.owasp.org/llm-top-10/

Stat 110
https://projects.iq.harvard.edu/stat110

Guides read for this edition

Tier-3 roadmap (Substack)
https://devshreebharatia.substack.com/p/tier-3-college-roadmap-for-software

Product companies from tier-3 (Medium)
https://medium.com/@hardiktaneja_99752/how-to-get-placed-in-product-based-companies-from-a-tier-3-college-f24bb4861b73

10 LPA from tier-3 (lets-code)
https://www.lets-code.co.in/blogs/how-to-get-a-10-lpa-job-as-a-fresher-from-a-tier-3-college/

HashedIn SDE-1 experience
https://leetcode.com/discuss/post/7620735/hashedin-by-deloitte-sde-1-interview-exp-urnm/

Off-campus backend internship
https://leetcode.com/discuss/post/7703521/title-sde-internship-interview-experienc-sbqj/

Intuit SDE intern (not selected)
https://www.geeksforgeeks.org/interview-experiences/intuit-interview-experience-for-sde-internship-off-campus/

Cyber jobs in India (talk)
https://sessionize.com/s/aseem-shrey/getting-a-cybersecurity-job-in-india/41330

Climb one rung at a time. The college name is a filter for the first round only — everything after it is the work in this guide.